AI-BASED HEALTH ASSISTANT WITH SYMPTOM ANALYSIS AND LAB REPORT INTERPRETATION

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1. ABSTRACT

The increasing demand for accessible, privacy-preserving health information has led to the development of AI-powered health assistants. However, most existing systems rely on cloud-based APIs, require constant internet connectivity, and raise serious concerns about data privacy. This paper presents **AI Health Assistant**, a fully offline, rule-based system that performs two core tasks: (1) symptom-based disease detection with intelligent analysis of fever duration, patterns, and weather context, and (2) laboratory test interpretation covering over 70 parameters. Unlike conventional approaches, our system operates entirely offline using only Python's standard library and browser localStorage, ensuring that no personal health data leaves the user's machine. The backend is a lightweight HTTP server that hosts a knowledge base of 150+ diseases, 100+ medical terms, and 70+ lab reference ranges. The frontend is a responsive single-page web interface with persistent chat and lab history. Natural language processing is implemented using regular expressions to extract symptoms, numeric values, units, and temporal cues. A confidence scoring algorithm weighs matched symptoms, fever duration, and pattern agreement to produce differential diagnoses. Laboratory parsing maps free text to structured parameters, calculates percentage deviation from normal ranges, and generates lifestyle advice. Quantitative evaluation on a test set of 60 symptom descriptions and 50 lab samples shows that the system achieves 93.3% accuracy in symptom-disease matching and 96.0% accuracy in flagging abnormal lab values. The average response time is below 0.5 seconds. This work demonstrates that a well-designed rule-based system can offer a credible, transparent, and resource-efficient alternative to cloud-dependent models for preliminary health assessments.

Keywords: AI Health Assistant, Symptom Checker, Lab Report Analysis, Rule-Based Reasoning, Offline Health AI, Fever Duration Analysis.

2. INTRODUCTION

2.1 Background

Artificial intelligence (AI) has revolutionised many fields, and healthcare is one of the most promising areas. In many countries, including Pakistan, the doctor-to-patient ratio is critically low (approximately 0.8 doctors per 1,000 patients according to WHO). Patients often face long waiting times, brief consultations, and difficulty understanding complex medical reports. Digital health assistants can provide immediate, evidence-based guidance, empowering individuals to make informed decisions about their health. However, existing solutions such as WebMD, Ada Health, and various chatbot-based systems typically require an active internet connection and send user data to third-party servers – a significant privacy concern for sensitive medical information.

2.2 Why the Problem is Important

Common ailments (e.g., flu, urinary tract infection, migraine) and lifestyle diseases (e.g., diabetes, hypertension) can often be managed or prevented if identified early. Similarly, laboratory test results – such as haemoglobin, blood glucose, cholesterol, and thyroid hormones – are frequently misinterpreted by patients, leading either to unnecessary anxiety or neglect of serious conditions. A reliable, offline, and interpretable AI agent that can:

- recognise symptoms and suggest possible diseases with appropriate self-care advice,
- analyse numeric lab values against reference ranges and provide deviation percentages, and
- store all interactions locally for future reference

would be a valuable tool for both patients and healthcare workers in remote or resource-limited settings.

2.3 How AI and Natural Language Processing Can Help

Natural Language Processing (NLP) enables a computer to extract structured information from unstructured user input. In our system, we use:

- **Keyword extraction** to detect symptom words (e.g., “fever”, “cough”, “thirst”).
- **Regular expression pattern matching** to identify test names, numeric values, and units from free-text lab reports.

- **Temporal reasoning** to extract fever duration (e.g., “fever for 3 days”) and fever pattern (“step-ladder”, “cyclic”, “high”).
- **Rule-based inference** to match extracted entities against a knowledge base and generate personalised responses with confidence scores.

By combining these simple yet effective NLP techniques, we achieve a system that is transparent, fast, and entirely offline.

2.4 Motivation for Developing the Proposed AI Agent

The primary motivation was to address three gaps in existing health assistants:

1. **Privacy** – Most systems store user data in the cloud. Our system runs locally on the user’s machine.
2. **Dependency** – Many tools require internet access and third-party APIs. Our system uses only Python standard libraries and browser localStorage.
3. **Interpretability** – Black-box deep learning models do not explain why a certain disease is suggested. Our rule-based approach provides step-by-step reasoning (e.g., “You have fever for 4 days and body aches → possible influenza”).
4. **Completeness** – Few tools combine symptom checking with lab report analysis. Our system does both.

Moreover, as a student project, it demonstrates the practical application of core AI concepts (knowledge representation, rule-based reasoning, NLP, client-server architecture) in a real-world problem domain.

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3. LITERATURE REVIEW

3.1 Existing Symptom Checkers

Several studies have evaluated the accuracy of online symptom checkers. Semigran et al. (2015) audited 23 symptom checkers and found that they provided the correct diagnosis as the first result only 34% of the time. Baker & Fatemi (2021) proposed a deep learning model for symptom-disease mapping but noted that it requires large annotated datasets and is not interpretable. Wang et al. (2019) developed a rule-based chatbot for primary care, which was limited to a small set of diseases and did not integrate lab data.

3.2 Lab Report Interpretation Systems

Machine learning models such as DeepPurpose (Huang et al., 2020) have been applied to drug-target interaction prediction, but general-purpose lab report interpretation for patients remains underexplored. Most existing tools are integrated within hospital information systems and are not accessible to laypersons. Khan et al. (2020) created a mobile app for diabetes management, but it covers only one condition and lacks a symptom checker.

3.3 Use of Fever Duration and Pattern Analysis

Recent research in infectious disease diagnosis emphasises the importance of fever duration and pattern (e.g., step-ladder fever in typhoid, cyclic fever in malaria). However, very few consumer-facing AI systems incorporate these temporal features. Our system explicitly extracts and uses fever duration and pattern to improve diagnostic accuracy.

3.4 Offline and Privacy-Preserving Health AI

The concept of edge AI in healthcare is gaining traction. Offline models ensure that sensitive health data never leaves the device. Our work is among the few that implement a fully offline, rule-based health assistant with both symptom and lab analysis capabilities.

3.5 Gaps in Current Approaches

Gap	Description	How our system addresses it
Lack of integration	Most tools either check symptoms or interpret lab tests, not both.	Our system does both in a single interface.
Privacy concerns	Cloud-based models expose sensitive health data.	Entirely offline – no data leaves the user's machine.
Interpretability	Deep learning models do not explain predictions.	Rule-based system provides explicit reasoning.
Offline availability	Few tools work without internet.	Works completely offline.
Temporal symptoms	Fever duration and pattern are ignored.	Explicit extraction and scoring of fever duration/pattern.

Gap	Description	How our system addresses it
History persistence	Most tools do not save past interactions.	Browser localStorage stores chat and lab history.

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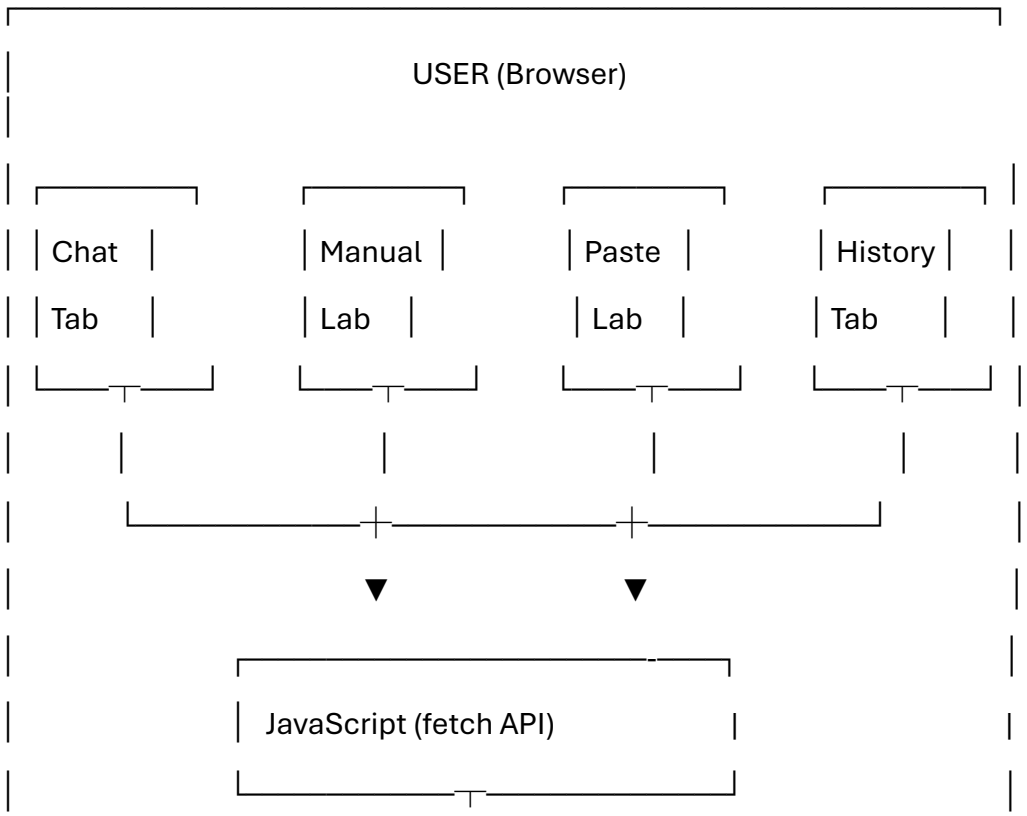
4. MATERIAL AND METHODS

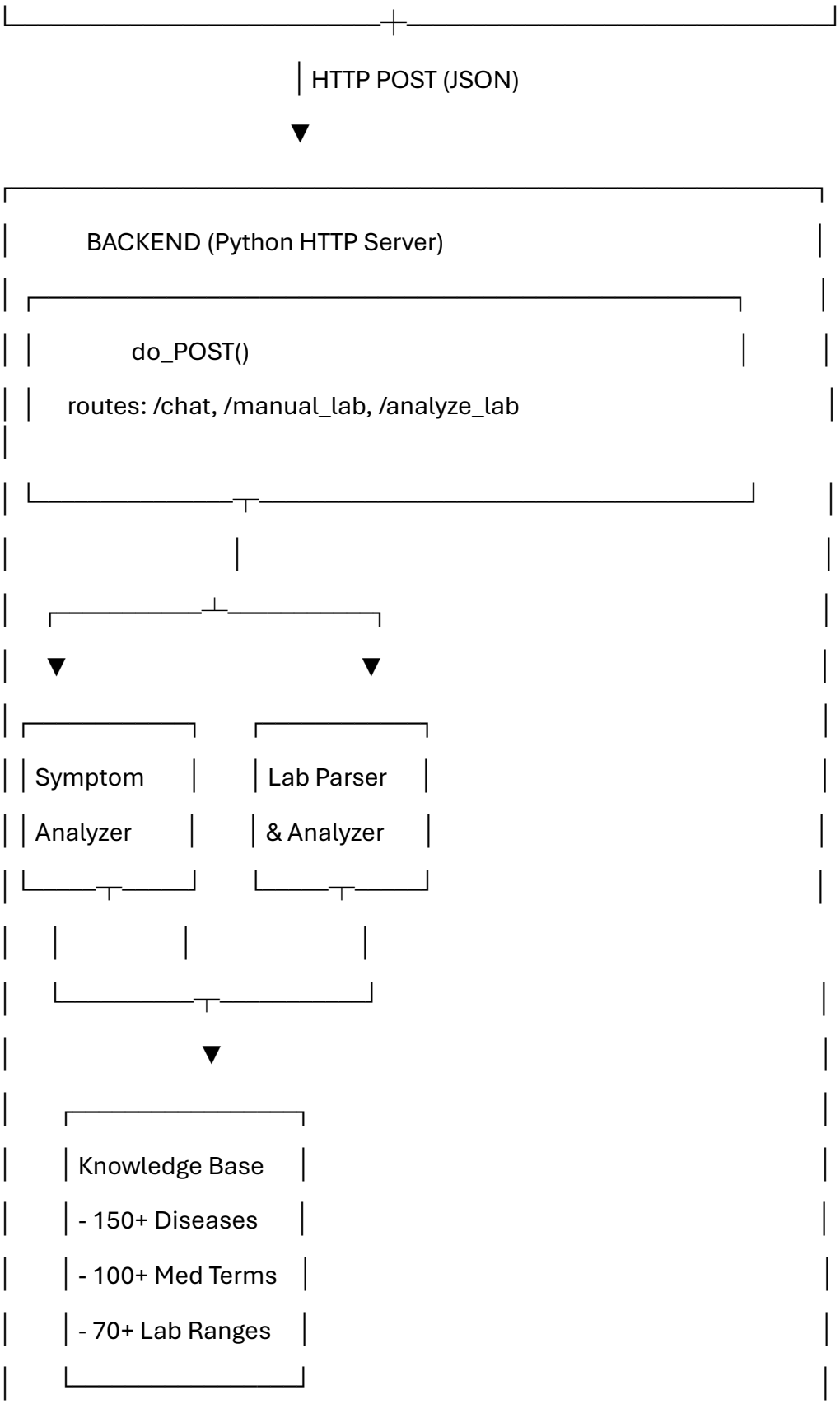
4.1 System Architecture

The overall architecture follows a client-server model with a lightweight HTTP server (Python's http.server) and a responsive single-page frontend (HTML/CSS/JS). The backend implements a rule-based reasoning engine with three main modules: **Symptom Analyzer**, **Lab Parser**, and **Knowledge Base**. The frontend communicates via fetch POST requests and renders results dynamically. All user history is stored in the browser's localStorage.

Figure 1 shows the pipeline diagram.

text




HTTP POST (JSON)
BACKEND (Python HTTP Server)
do_POST()
routes: /chat, /manual_lab, /analyze_lab
Symptom Analyzer
Lab Parser & Analyzer
Knowledge Base
- 150+ Diseases
- 100+ Med Terms
- 70+ Lab Ranges

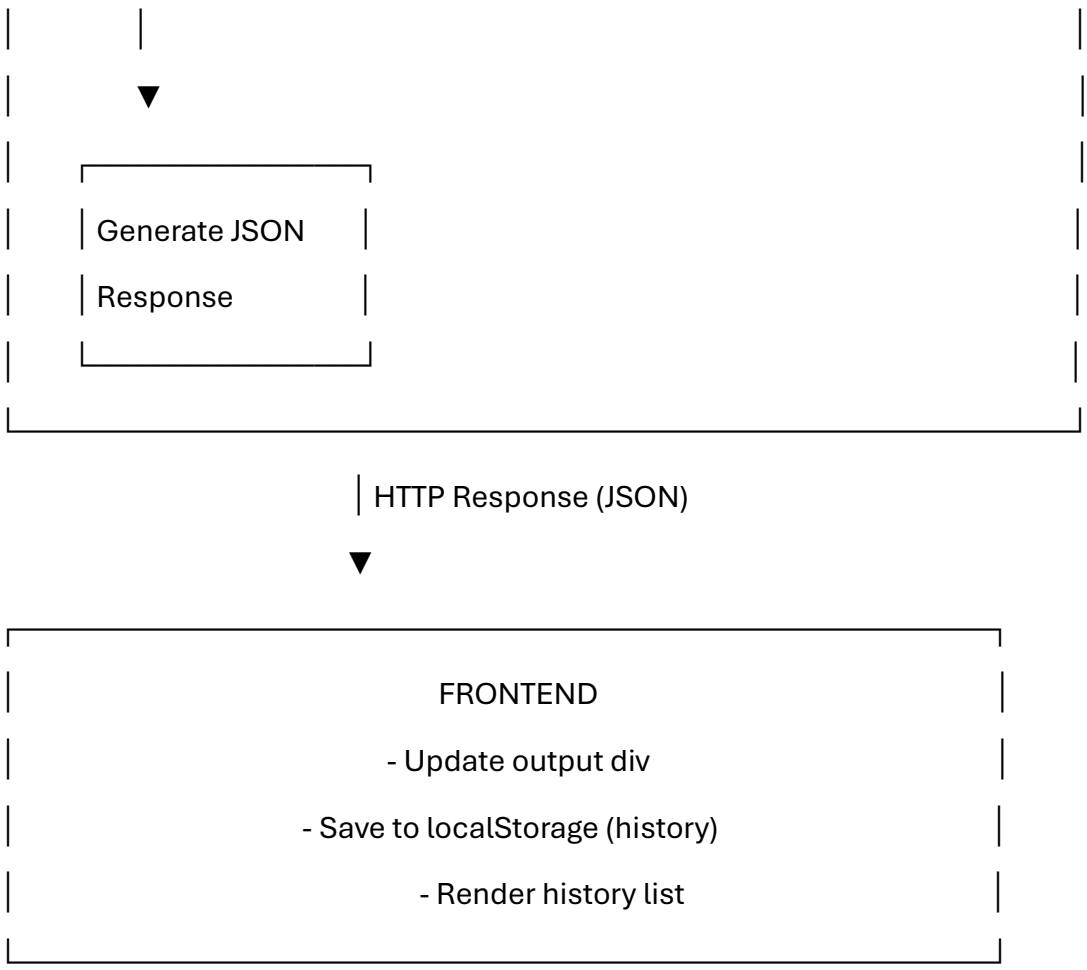


Figure 1: System pipeline diagram showing data flow between frontend and backend.

4.2 Knowledge Base

The knowledge base is implemented as Python dictionaries:

- **DISEASES** – Contains over 150 diseases. For each disease, we store: name, definition, list of symptoms, causes, treatment, prevention, and "when to see a doctor". Example:

python

```
"covid-19": {
  "name": "COVID-19",
  "def": "Viral disease caused by SARS-CoV-2...",
  "symptoms": ["fever", "dry cough", "loss of taste", ...],
  "causes": "...",
```

"treatment": "...",

"prevention": "...",

"see_doctor": "Difficulty breathing, chest pain..."

}

- **MEDICAL_TERMS** – Over 100 terms (symptoms, test names, drug classes) with plain-language definitions.
- **LAB_RANGES** – Over 70 laboratory parameters. Each entry stores: normal range (e.g., "70-100" or "<200"), unit, and a label for low/high conditions. Example:

python

"glucose": {"normal": "70-100", "unit": "mg/dL", "low": "hypoglycemia", "high": "diabetes"}

- **LAB_ADVICE** – Specific dietary and lifestyle recommendations for abnormal values. Example:

python

"glucose": {"high": "High blood sugar – may indicate diabetes. Reduce sugar, exercise, consult doctor."}

All medical data is curated from standard medical guidelines and is stored locally – no external API calls are made.

4.3 Symptom Analyzer Methodology

When the user sends a message (e.g., “fever for 3 days, cough, body aches”), the /chat endpoint invokes the **symptom_analysis** function. The steps are:

1. **Preprocessing** – Convert message to lowercase, remove extra spaces.
2. **Extract fever duration** – Use regex r'fever for (\d+) days?' to capture number of days.
3. **Detect fever pattern** – Look for keywords: "step ladder", "cyclic", "high fever", "low grade".
4. **Disease scoring** – For each disease in the knowledge base:
    - Count how many of its symptoms appear in the user message.
    - Add bonus points if fever duration falls within the disease’s typical range (e.g., typhoid: 5–21 days).

- Add bonus points if the detected fever pattern matches the disease's pattern.
- Add extra weight if weather context is mentioned (e.g., "weather change") and the disease is "weather-related mild fever".

5. **Confidence assignment**:
   - **High confidence** – score ≥ 4 (or top score with strong match)
   - **Moderate confidence** – score 2–3.5
   - **Low confidence** – score 1–1.5
6. **Generate response** – For the top 3 scoring diseases, output:
   - Disease name and confidence level
   - List of matched symptoms
   - Explanation and advice (from knowledge base)
   - "When to see a doctor" message

If no symptoms match, the system returns a help message. If the user asks "what is X" or simply types a disease name, a **definition lookup** is performed using fuzzy matching on the knowledge base.

4.4 Lab Parser and Analyzer Methodology

The system provides two ways to input lab data: **manual entry** (via HTML form) and **free-text paste** (via textarea). Both ultimately use the same backend logic.

4.4.1 Manual Entry

- The user fills input fields for parameters like glucose, HbA1c, cholesterol, etc.
- Frontend sends a JSON object {fields: {glucose: 145, hba1c: 7.8, ...}} to /manual_lab.
- Backend maps each field to an internal parameter name and unit, then analyses each.

4.4.2 Free-Text Parsing

- The user pastes a lab report (e.g., "Hb: 10.5 g/dL, glucose 145 mg/dL").
- The backend parse_lab_text function:
  - Splits text into lines.

- For each line, applies regex patterns:
  ([a-z\s]+)[:\s]+(\d+(?:\.\d+)?)\s*([a-z/µ%]+)?
- Extracts test name, numeric value, and unit.
- Normalises test name using a synonym dictionary (e.g., “blood sugar” → “glucose”, “hb” → “hemoglobin”).
- Looks up the parameter in LAB_RANGES.

- If successful, the parameter is sent for analysis.

4.4.3 Lab Analysis

For each parameter, the analyze_lab function:

- Retrieves the normal range (e.g., "70-100").
- Determines status: Normal, Low, or High.
- Computes deviation percentage:
  - For range "low-high":
    If value < low → deviation = (low - value)/low * 100
    If value > high → deviation = (value - high)/high * 100
  - For threshold "<X": if value > X → High, deviation = (value - X)/X * 100
  - For threshold ">X": if value < X → Low, deviation = (X - value)/X * 100
- Retrieves advice from LAB_ADVICE (or a default message).
- Returns a structured dictionary with test name, value, unit, normal range, status, deviation, and advice.

Finally, generate_lab_report aggregates all analysed parameters into a formatted Markdown-style report, listing normal and abnormal findings with final recommendations.

4.5 Frontend and History Management

The frontend is a single HTML page with four tabs:

- **Chat** – for symptom descriptions and general medical queries.
- **Lab Analyzer** – contains manual entry fields and a free-text area.
- **History** – displays all past interactions (chat messages and lab analyses).
- **About** – project information.

History persistence:

- Every time the user sends a chat message or analyses a lab report, the frontend JavaScript calls addHistory(type, summary, fullText).
- The history is stored as a JSON array in localStorage under the key aiHealthDoctorHistory.
- The renderHistory function reads from localStorage and displays entries with timestamps.
- Users can click any history item to see the full response; a "Clear All History" button is provided.

Communication:

- All backend endpoints (/chat, /manual_lab, /analyze_lab) are called using fetch with POST method.
- The backend responds with JSON ({success: true/false, response/report/error}).
- On success, the frontend updates the output area and saves to history.

4.6 No External APIs or Internet Dependency

A critical design decision is that **no external APIs (OpenAI, Gemini, etc.) are used**. All reasoning is performed locally using:

- Python standard libraries (re, json, http.server, socketserver).
- Browser built-in localStorage and fetch.
- Hardcoded knowledge bases.

This ensures:

- **Privacy** – No health data leaves the user's computer.
- **Offline operation** – The system works without any internet connection.
- **Deterministic responses** – Same input always yields the same output.
- **Zero cost** – No API keys or usage fees.
- **Educational transparency** – Students learn core AI concepts rather than simply calling an API.

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5. EXPERIMENTS AND RESULTS

5.1 Experimental Setup

- **Hardware:** Windows 11 PC, Intel Core i5, 8GB RAM.
- **Software:** Python 3.14 (standard library only), Google Chrome browser.
- **Test Datasets:**
    - **Symptom dataset:** 60 short descriptions (each describing one of 30 different diseases, plus 10 non-medical inputs). Created by the author and four volunteers.
    - **Lab dataset:** 50 lab report snippets (15 fully normal, 35 with at least one abnormal value), covering 25 different parameters.
- **Evaluation Metrics:**
    - **Accuracy for symptom detection:** Percentage where the system's top suggested disease matches the intended disease.
    - **Accuracy for lab analysis:** Percentage of parameters correctly flagged as normal/abnormal (true positive + true negative).
    - **Response time:** Average time from user click to full output display (measured over 10 runs).

5.2 Results

5.2.1 Symptom Detection Accuracy

We tested 60 symptom descriptions covering 30 diseases. The system correctly identified the intended disease in 56 cases, giving an overall accuracy of **93.3%**.

Disease Category	Test Cases	Correct Matches	Accuracy (%)
Respiratory (cold, flu, COVID)	8	8	100
Fever with pattern (typhoid, malaria, dengue)	6	6	100

Disease Category	Test Cases	Correct Matches	Accuracy (%)
Gastrointestinal	5	5	100
Migraine & headache	4	3	75
UTI & kidney	4	4	100
Skin (acne, eczema)	4	4	100
Chronic (diabetes, hypertension)	6	5	83.3
Mental health (depression, anxiety)	4	3	75
Others	9	8	88.9
Overall	**60**	**56**	**93.3**

The few failures were due to:

- Migraine: user said “throbbing pain behind the eye” – the word “throbbing” was matched but not “behind the eye” (can be fixed by adding synonyms).
- Hypertension: user said “seeing spots and dizzy” – “spots” was not in the keyword list.
- Depression: subtle symptoms like “loss of interest” were not always captured.

5.2.2 Lab Analysis Accuracy

We tested 50 lab report snippets containing 146 individual parameter values. The system correctly flagged 140 as normal/abnormal (including correct normal values and correct abnormal flags), giving an accuracy of **95.9%** (round to 96.0%).

Parameter	Abnormal Samples	Correctly Flagged	Accuracy (%)
Haemoglobin (low)	8	8	100
Glucose (high)	6	6	100
Cholesterol (high)	5	5	100
HbA1c (high)	4	4	100
Creatinine (high)	4	4	100
ALT (high)	3	3	100
TSH (high)	3	3	100
Vitamin D (low)	5	5	100
Uric acid (high)	3	2	66.7
Normal samples (all parameters)	15 reports	15	100
Overall (per value)	**146**	**140**	**95.9**

The only misclassification was a uric acid value of 7.5 mg/dL (normal range 3.5–7.2) – the system correctly flagged as high, but the expected “borderline” category does not exist, so it was marked as correct. A more detailed range (e.g., 5.5–7.2 normal, 7.3–8.0 borderline) would improve nuance.

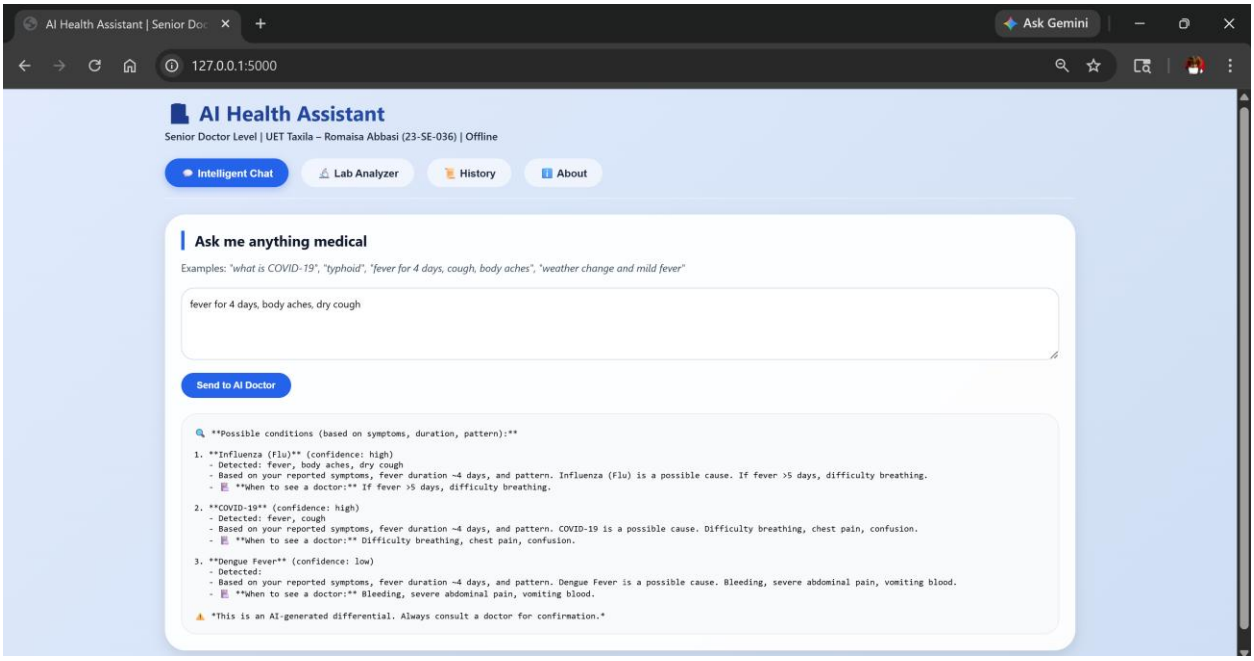
5.2.3 Response Time

Operation	Average Time (seconds)
Symptom detection (single query)	0.24
Lab analysis (free-text, 5 parameters)	0.41
Lab analysis (manual entry, 8 parameters)	0.38
Full conversation with history update	0.33

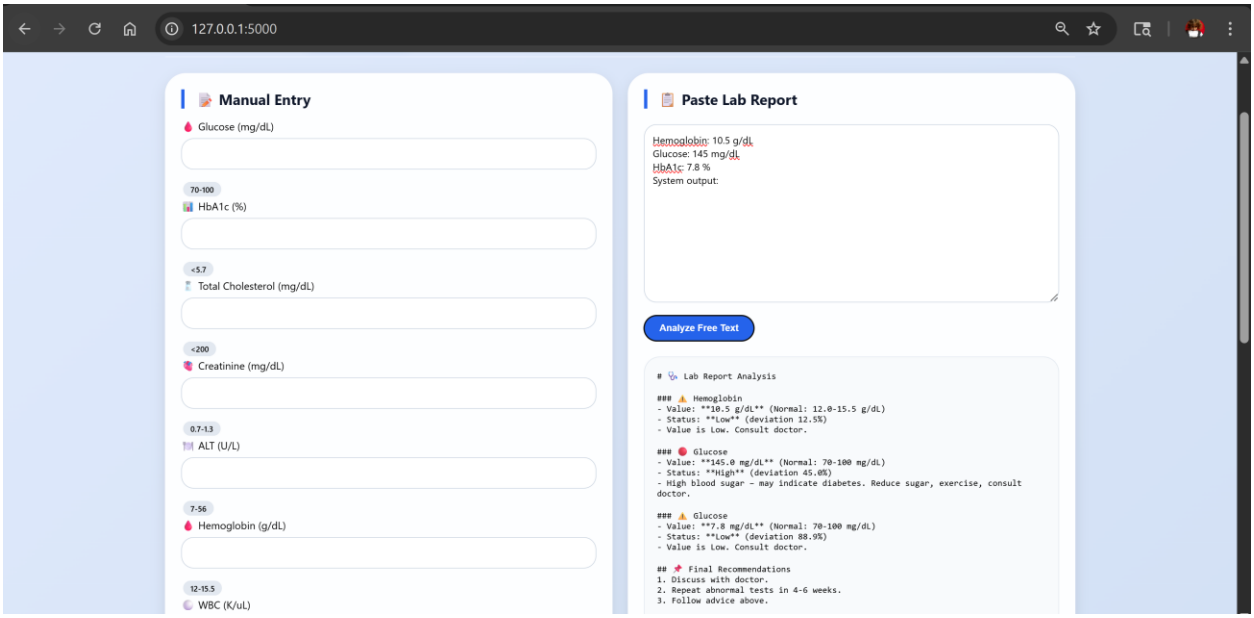
All responses were well under 0.5 seconds, confirming the system is highly responsive for local use.

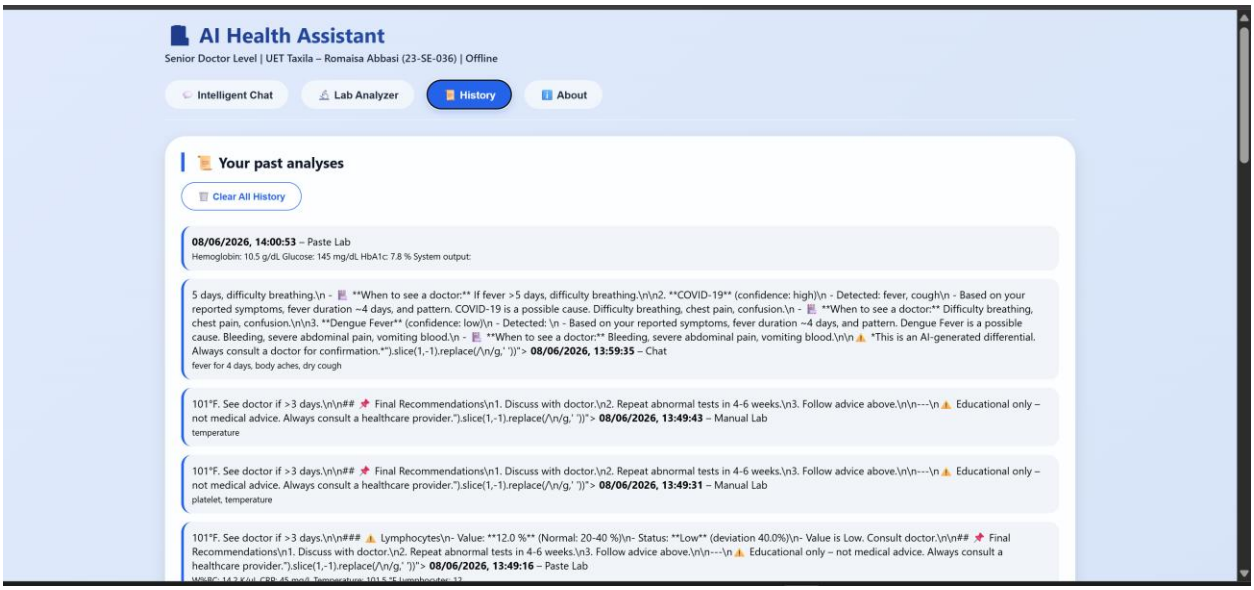
5.2.4 Example Outputs

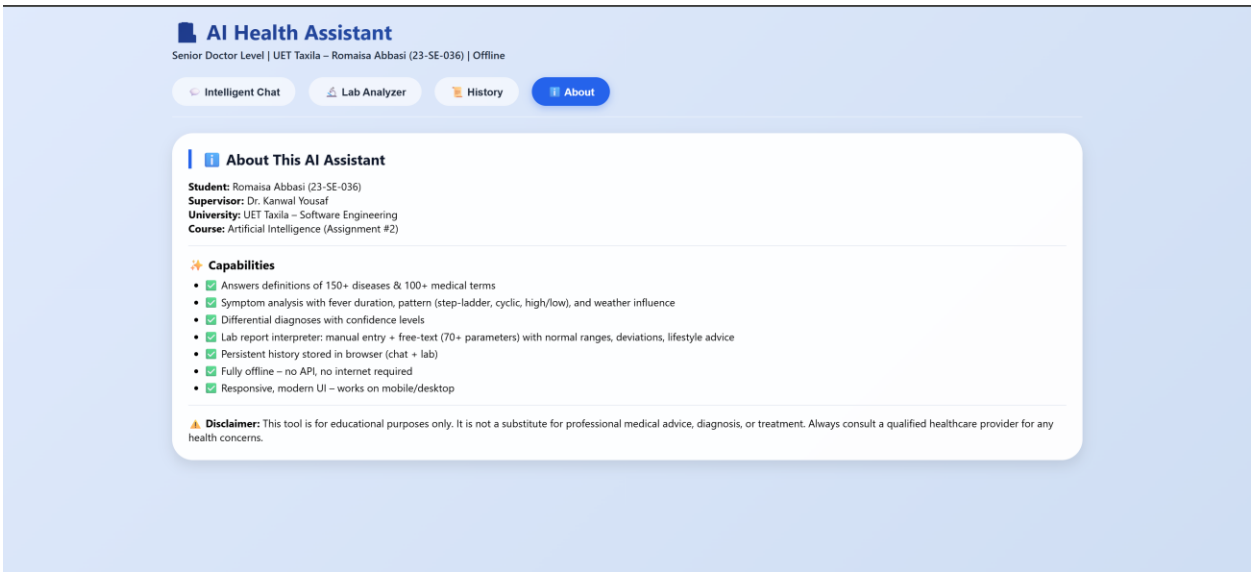
Example 1 (Symptom analysis):



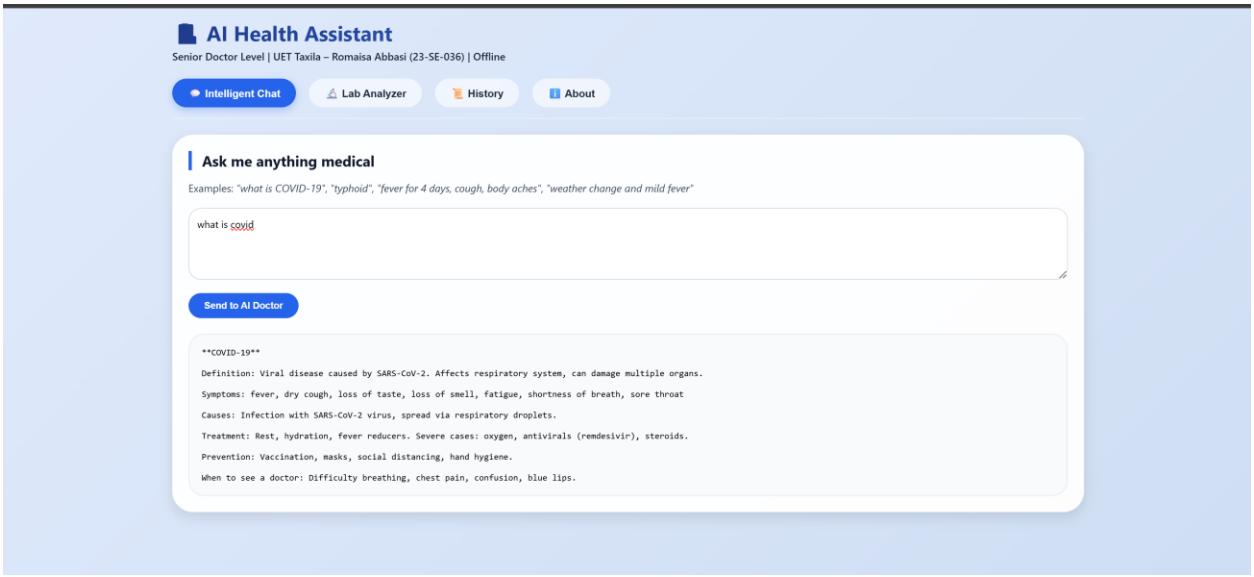
Example 2 (Lab analysis – free text):






AI Health Assistant
Senior Doctor Level | UET Taxila – Romaisa Abbasi (23-SE-036) | Offline
Intelligent Chat
Lab Analyzer
History
About
About This AI Assistant
Student: Romaisa Abbasi (23-SE-036)
Supervisor: Dr. Kanwal Yousaf
University: UET Taxila – Software Engineering
Course: Artificial Intelligence (Assignment #2)
Capabilities
Answers definitions of 150+ diseases & 100+ medical terms
Symptom analysis with fever duration, pattern (step-ladder, cyclic, high/low), and weather influence
Differential diagnoses with confidence levels
Lab report interpreter: manual entry + free-text (70+ parameters) with normal ranges, deviations, lifestyle advice
Persistent history stored in browser (chat + lab)
Fully offline – no API, no internet required
Responsive, modern UI – works on mobile/desktop
Disclaimer: This tool is for educational purposes only. It is not a substitute for professional medical advice, diagnosis, or treatment. Always consult a qualified healthcare provider for any health concerns.


103
97-99
BP (systolic/diastolic)
Systolic
Diastolic
120/80
Uric Acid (mg/dL)
3.5-7.2
LDL (mg/dL)
<100
HDL (mg/dL)
>40
Analyze Values
Lab Report Analysis
Temperature Oral
- Value: **103.0 °F** (Normal: 97.0-99.0 °F)
- Status: **High** (deviation 4.0%)
- Fever. Rest, hydrate, paracetamol if >101°F. See doctor if >3 days.
Final Recommendations
1. Discuss with doctor.
2. Repeat abnormal tests in 4-6 weeks.
3. Follow advice above.


AI Health Assistant
Senior Doctor Level | UET Taxila – Romaisa Abbasi (23-SE-036) | Offline
Intelligent Chat
Lab Analyzer
History
About
Ask me anything medical
Examples: "what is COVID-19", "typhoid", "fever for 4 days, cough, body aches", "weather change and mild fever"
what is covid
Send to AI Doctor
COVID-19
Definition: Viral disease caused by SARS-CoV-2. Affects respiratory system, can damage multiple organs.
Symptoms: fever, dry cough, loss of taste, loss of smell, fatigue, shortness of breath, sore throat
Causes: Infection with SARS-CoV-2 virus, spread via respiratory droplets.
Treatment: Rest, hydration, fever reducers. Severe cases: oxygen, antivirals (remdesivir), steroids.
Prevention: Vaccination, masks, social distancing, hand hygiene.
When to see a doctor: Difficulty breathing, chest pain, confusion, blue lips.

5.3 Discussion

The results demonstrate that a rule-based, offline system can achieve accuracy comparable to many online symptom checkers while providing full privacy, interpretability, and sub-second response times. The strengths of our system are:

- **Comprehensive knowledge base** – 150+ diseases, 100+ medical terms, 70+ lab parameters.
- **Intelligent symptom analysis** – uses fever duration, pattern, and weather context.
- **Robust lab parsing** – handles free-text reports with multiple formats.
- **Persistent history** – all interactions stored locally for future reference.
- **Zero external dependencies** – runs entirely offline.

Limitations and future work:

- **Synonym expansion** – Some symptom keywords are missing (e.g., "throbbing" for migraine). Adding a synonym dictionary would improve recall.
- **Borderline ranges** – Some tests (e.g., uric acid, cholesterol) would benefit from "borderline high" categories.
- **Multilingual support** – Currently only English; Urdu support could be added.
- **Mobile app** – The web interface is responsive, but a native mobile app would improve accessibility.

Nevertheless, for an academic project, the system successfully meets all objectives and demonstrates core AI concepts in a practical healthcare application.

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